today: set up

monday: grover's algorithm

unstructured search over N items $\to 2^n$

from $O(N)$ to $O(\sqrt{N})$ time

brute force $O(2^n)$ to $O(\sqrt{2^n})$

$= O((2^n)^{1/2}) = O(2^{n/2})$

wednesday: shor's algorithm

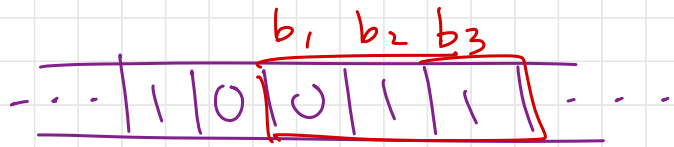
factor n bit N in $O(n^3)$

What is computation?

any evolution of state by a sequence of simple, local steps

classical computing

state: $x \in \{0,1\}^n$ TM tape



evolve: logical operators

eg Tofolli gate $b_1 b_2 b_3 \to b_1 b_2 (b_1 \wedge b_2 \oplus b_3)$

$011 \to 01(0 \wedge 1 \oplus 1) \to 01(0 \oplus 1) \to 011$

output: final state

implementation:

physical switch	pre 1940
vacuum tube	1940s-50s
transistors	50s onward

Probabilistic computing

ops can be probilistic

evolve eg:

"with prob. 1/4, apply Tofolli
with prob. 3/4, do nothing"

state: random variable $p_x \in \mathbb{R}$, $x \in \{0,1\}^n$

$$p_x \geq 0, \quad \sum_{x \in \{0,1\}^n} p_x = 1$$

sort of have access to 2^n states?

ex

Simple ProbAlg:

initialize 3 bits 111
with prob 1/2, ¬ first bit

x	p_x
111	1/2
011	1/2
000	0
⋮	

$x \in \{0,1\}^3$

output: sample from distribution p_x

implementation:

simulation

physical randomness

Quantum computing

state: linear combos of amplitudes

$x \in \{0,1\}^n$ $\quad \alpha_x \in \mathbb{C}$ technically

$\in \mathbb{R}$

states: n qubits

$x = |01101\rangle$ "ket"

5 qubits

amplitudes unitary

$$\sum_{x \in \{0,1\}^n} \alpha_x^2 = 1$$

evolve via unitary operation

unitary $\alpha_x \rightarrow f(\alpha_x)$ unitary

reversible : has an undo

output: sample from prob dist :

$$p_x = \alpha_x^2$$

"extract all"

ex First Quantum Alg

initialize 1 qubit $A = |0\rangle$

$\neg A$

extract all

$|0\rangle = 1$
$|1\rangle = 0$

$|0\rangle = 0$
$|1\rangle = 1$

$P_0 = 0^2 = 0$

$P_1 = 1^2 = 1$

$|0\rangle = 0$
$|1\rangle = 1$

Unitary prop

$1^2 + 0^2 = 1$ ✓

$0^2 + 1^2 = 1$ ✓